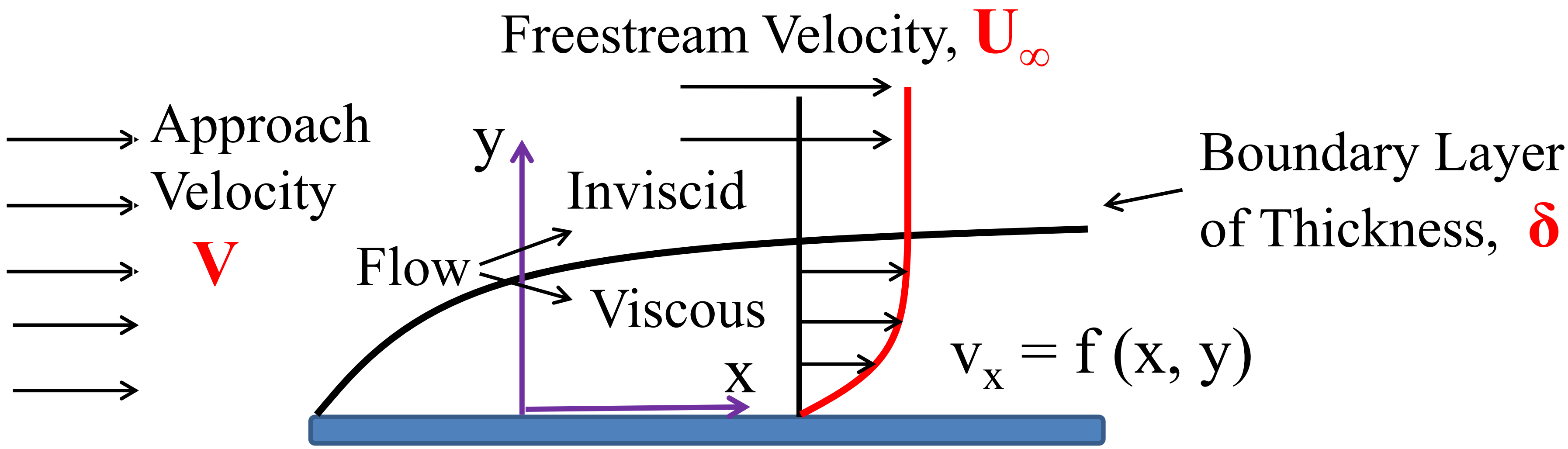
External Incompressible Viscous Flow – Boundary Layer



Flow over a flat plate

$v_x = f\,(x, y)$ Viscous 2D flow inside BL

$v_x = U_\infty$ Inviscid flow outside BL

$v_x = 0.99U_\infty$
at $y = \delta$

δ - boundary layer thickness

Integral Equation

$$\frac{dN}{dt}\Big|_{system} = \frac{\partial}{\partial t}\int_{CV} \eta\, \rho d\forall + \int_{CS} \eta\, \rho \overrightarrow{V}.\overrightarrow{dA}$$

Displacement and Momentum Thicknesses

$$\delta^* = \int_0^\delta \left(1 - \frac{v_x}{U}\right) dy, \qquad \theta = \int_0^\delta \frac{v_x}{U}\left(1 - \frac{v_x}{U}\right) dy$$

MI Equation

$$\boldsymbol{\frac{\tau_w}{\rho} = \frac{d}{dx}\left(U^2\theta\right) + \delta^* U \frac{dU}{dx}}$$

Turbulent Flow

TURBULENT FLOW

FULLY DEV STEADY FLOW

$$0 = -\frac{\partial P}{\partial z} + \frac{1}{r}\frac{\partial}{\partial r}(r\,\tau_{rz}).$$

Z COMP. OF EQ^N OF MOTION

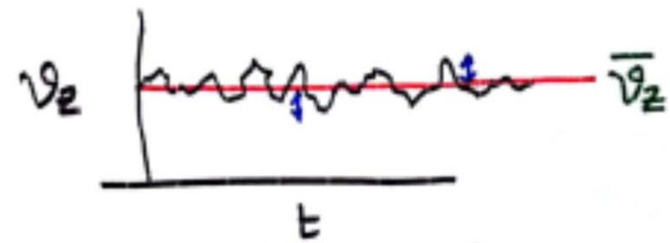
$$\tau_{rz} = \frac{r}{2}\frac{\partial P}{\partial z} + \boxed{C}$$

AT $r=0$ $\tau_{rz}=0 \Rightarrow C=0$

$$\tau_{rz} = \frac{r}{2}\frac{\partial P}{\partial z}$$

$$\tau_W = -\tau_{rz}\Big|_{r=R} = -\frac{R}{2}\frac{\partial P}{\partial z}$$

VALID BOTH FOR LAM. & TURB FLOW ✓



$$v_z = \bar{v}_z + v_z'$$

INSTANTANEOUS VEL. (v_z)

TIME SMOOTHED VEL. ($\bar{v}_z$)

FLUCTUATING COMPONENT OF VEL. (v_z')

$$\overline{v_z'} = 0 \qquad \overline{(v_z')^2} \neq 0$$

$$\frac{\sqrt{\overline{(v_z')^2}}}{\langle \bar{v}_z \rangle} = \text{MEASURE OF TURBULENCE}$$

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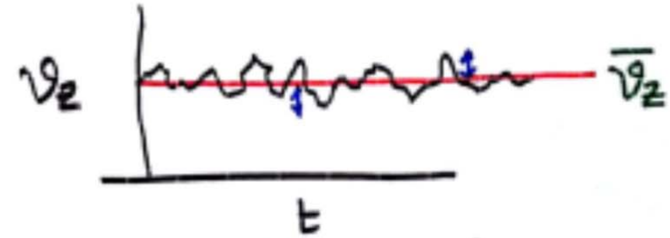


$$v_z = \bar{v}_z + v'_z$$

- v_z — INSTANTANEOUS VEL.
- $\bar{v}_z$ — TIME SMOOTHED VEL.
- v'_z — FLUCTUATING COMPONENT OF VEL.

$$\overline{v'_z} = 0 \qquad \overline{(v'_z)^2} \neq 0$$

$$\frac{\sqrt{\overline{(v'_z)^2}}}{\langle \bar{v}_z \rangle} = \text{MEASURE OF TURBULENCE}$$

Turbulent Flow Contd

Using in NS equation

x COMP.

$$\frac{\partial}{\partial t}\rho(\bar{v}_x + \cancel{v_x'}) = -\frac{\partial}{\partial x}(\bar{p} + \cancel{p'}) - \left[\frac{\partial}{\partial x}\rho(\bar{v}_x + v_x')(\bar{v}_x + v_x') + \frac{\partial}{\partial y}\rho(\bar{v}_y + v_y')(\bar{v}_x + v_x') + \frac{\partial}{\partial z}(\bar{v}_z + v_z')(\bar{v}_x + v_x')\rho\right] + \mu\nabla^2(\bar{v}_x + v_x') + \rho g_x.$$

$\overline{v_x' v_y'} \neq 0$

$\overline{v_x'} = 0 \quad \overline{v_y'} = 0$

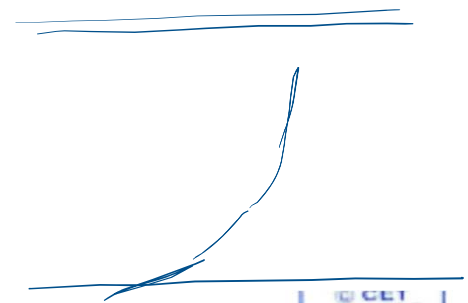
TAKE TIME AVERAGE

$$\frac{\partial}{\partial t}\rho\bar{v}_x = -\frac{\partial \bar{p}}{\partial x} - \left(\frac{\partial}{\partial x}\rho\bar{v}_x\bar{v}_x + \frac{\partial}{\partial y}\rho\bar{v}_y\bar{v}_x + \frac{\partial}{\partial z}\rho\bar{v}_z\bar{v}_x\right) + \mu\nabla^2\bar{v}_x + \rho g_x - \left(\frac{\partial}{\partial x}\rho\overline{v_x' v_x'} + \frac{\partial}{\partial y}\rho\overline{v_x' v_y'} + \frac{\partial}{\partial z}\rho\overline{v_x' v_z'}\right)$$

REYNOLD'S STRESSES

Turbulent Flow Contd

Universal velocity profile



VISCOUS SUB LAYER

$$v_z^+ \equiv \frac{\bar{v}_z}{v_*} = \frac{y v_*}{\nu} = y^+ \quad \boxed{0 \le y^+ \le 5}$$

$$v_* = \sqrt{\frac{\tau_w}{\rho}} = \text{FRICTION VEL}$$

TRANSITION REGION

$$\frac{\bar{v}_z}{v_*} = v_z^+ = 2.5 \ln \frac{y v_*}{\nu} + 5.0, \quad \boxed{5 < y^+ < 26}$$

TURB. CORE

$$v_z^+ = \frac{1}{0.36} \ln y^+ + 3.8, \quad \boxed{y^+ \ge 26}$$

Turbulent Flow Contd

Universal velocity profile

POWER LAW EQ^N EMPIRICAL

$$\frac{\bar{v}_z}{U} = \left(\frac{y}{R}\right)^{1/7}$$

CENTRELINE VEL

DIST. FROM THE PIPE WALL

RAD.

$\boxed{Re \sim 10^4 - 10^5}$

1/7 th POWER LAW PROFILE

$$\frac{\bar{V}}{U} = \frac{2n^2}{(n+1)(2n+1)}$$

$n \sim 1/7$ $\quad \frac{\bar{V}}{U} = 0.8$ | 0.5

$\frac{y}{R} < 0.04$ → INFINITE VEL. GRAD AT THE WALL

DOES NOT GIVE ZERO SLOPE AT THE CENTRE LINE

$$\frac{\tau_w}{\rho} = U^2 \frac{d\delta}{dx}\int_0^1 \frac{u}{U}\left(1-\frac{u}{U}\right)d\eta$$

Turbulent Boundary Layers

BOUNDARY LAYERS IN TURBULENT FLOW

✓ $\dfrac{v_x}{U} = \left(\dfrac{y}{\delta}\right)^{1/7} = \eta^{1/7}$

M I EQN: $\dfrac{\tau_w}{\rho} = U^2 \dfrac{d\theta}{dx}$

$$\tau_w = -\frac{R}{2}\frac{\partial p}{\partial z}.$$

$$\frac{p_1 - p_2}{\rho} \equiv \frac{\Delta P}{\rho} = h$$ (h = HEAD LOSS FOR A HORIZONTAL PIPE)

$$h = f\,\frac{L}{D}\,\frac{\bar{V}^2}{2}$$ ($\bar{V}$ = AV. FLOW VEL.)

$$\tau_w = -\frac{R}{2}\frac{\partial p}{\partial z} = +\frac{R}{2}\frac{\Delta P}{L} = \frac{R}{2L}\rho h.$$

$$\tau_w = \frac{R}{2L}\rho f \frac{L}{D}\frac{\bar{V}^2}{2} = \frac{1}{8}\rho f \bar{V}^2$$

BLASIUS CORRELATION

$$f = \frac{0.3164}{Re^{0.25}}$$

$$\tau_w = 0.03325\,\rho \bar{V}^2 \left(\frac{\nu}{R\bar{V}}\right)^{0.25}$$

USING 1/7th POWER LAW

$$\boxed{\tau_w = 0.0225\,\rho U^2 \left(\frac{\nu}{U\delta}\right)^{1/4}}$$

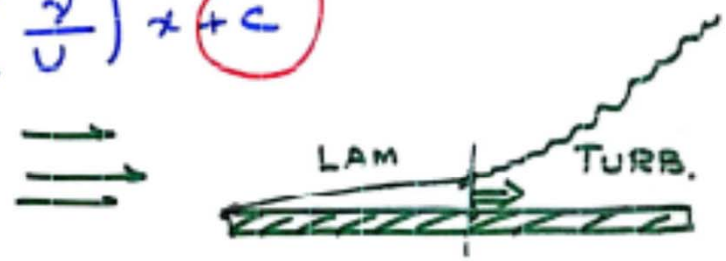
Turbulent Boundary Layers

MI eqn.

$$0.0225\left(\frac{\nu}{U\delta}\right)^{1/4} = \frac{d\delta}{dx}\int_0^1 \eta^{1/7}\left(1-\eta^{1/7}\right)d\eta = \frac{7}{72}\frac{d\delta}{dx}$$

$$\rightarrow \frac{4}{5}\delta^{5/4} = 0.23\left(\frac{\nu}{U}\right)^{1/4}x + c$$

BC $\boxed{x=0 \qquad \delta=0} \Rightarrow c = 0.$



$$\boxed{\delta = 0.37/(Re_x)^{1/5}}$$

Re

$$\boxed{c_f = \frac{\tau_w}{\frac{1}{2}\rho U^2} = 0.045\left(\frac{\nu}{U\delta}\right)^{1/4} = \frac{0.0577}{(Re_x)^{1/5}}}$$

$$5\times10^5 < Re_x < 10^7$$

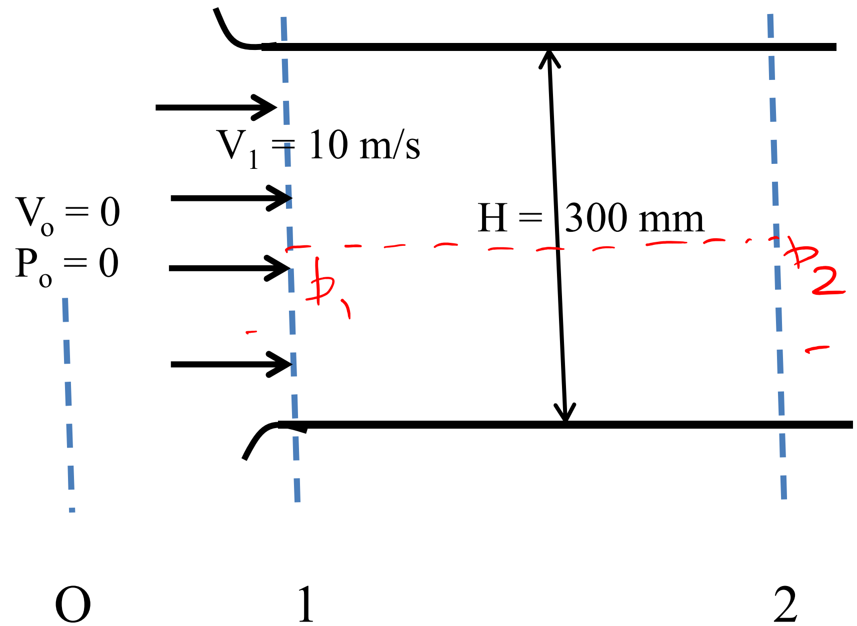
Flow of air (= 123 kg/m3) develops in a flat horizontal duct following a well rounded entrance section. The duct height is H = 300 mm. Turbulent boundary layer grows on the duct walls, but the flow is not yet fully developed. Assume that the velocity profile in each boundary layer is given by $u/U = (y/\delta)^{1/7}$. The inlet flow is uniform with V = 10 m/s at section 1. At section 2, the boundary layer thickness on each wall of the channel is δ_2 = 100 mm.

(a) Show that for this flow, $\delta^* = \delta/8$.

(b) Evaluate the static gage pressure at section 2

(c) Find the average wall shear stress between the entrance and section 2 at L = 5 m.

The region outside the entrance (location O in the figure) an be taken to be still air with pressure equal to the atmospheric pressure



$\delta^* = \delta/8$

$$\delta^* = \delta \int_0^1 \cancel{\left(1 - \frac{u}{U}\right)} d\eta = \delta \int_0^1 \left(1 - \eta^{1/7}\right) d\eta$$

$$\frac{\delta^*}{\delta} = \frac{1}{8}$$

$$V_1 A_1 = V_1 W H = V_2 A_2$$

$$V_1 A_1 = V_2 W \left(H - 2\delta_2^*\right)$$

$$V_2 = 10.9 \text{ m/s}$$

$$\frac{p_0}{\rho} + \cancel{\frac{v_0^2}{2}} = \frac{p}{\rho} + \frac{v^2}{2}$$

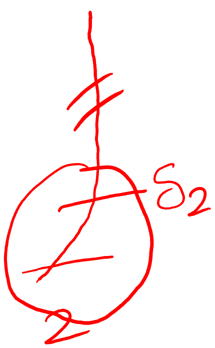
$$p_{1g} = p_1 - p_0 = -\frac{1}{2}\rho v_1^2 = -6105 \text{ Pa}$$

$$p_{2g} \rightarrow -73 \text{ Pa}.$$

Apply M^2 eq^n

$$F_{sx} = \int_{CS} u \rho \vec{V} \cdot d\vec{A}$$

$$(p_1 - p_2) w \frac{H}{2} - \cancel{\tau w L} = V_1 \left\{ -\rho V_1 \frac{H}{2} w \right\}$$



$$+ v_2 \left\{ \rho v_2 \left(\frac{H}{2} - S_2 \right) w \right\}$$

$$+ \int_0^{S_2} u \rho u w \, dy = A$$

$$A = \rho V_2^2 \delta_2 w \int_0^1 \eta^{2/7} \, d\eta$$

$1/7^{th}$ Power Law

$$A = \rho V_2^2 \frac{7}{9} \delta_2 w$$

$$\tau = 0.3 \; N/m^2$$